Technical review

Item #3 · future-exif.jpg · Run ID: 20261113T232757Z-cbb6f3

Path	`R:\evidence\images\future-exif.jpg`
Size (bytes)	120443
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.jpg
Detected type (signature)	JPEG image
Status	No issues found
Case	RV-SAMPLE-001
Evidence ID	E0013
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`88e5924ddfef6952f1fab164adbb9b04bfb21a593dfdf3d949022fb33577811c`
md5	`2e6a144de59544131c1f2ef04735cf27`

Findings

[Informational] Non-standard compression tables (QT_NON_STANDARD)

What it shows	The quantization tables are not IJG standard tables; the nearest IJG quality is 90 (distance 578).
What it does not prove	Cameras and many apps use their own tables; a quality figure for them is only an approximation and is not stated as fact.

[Informational] Compression tables not in the database (QT_UNKNOWN)

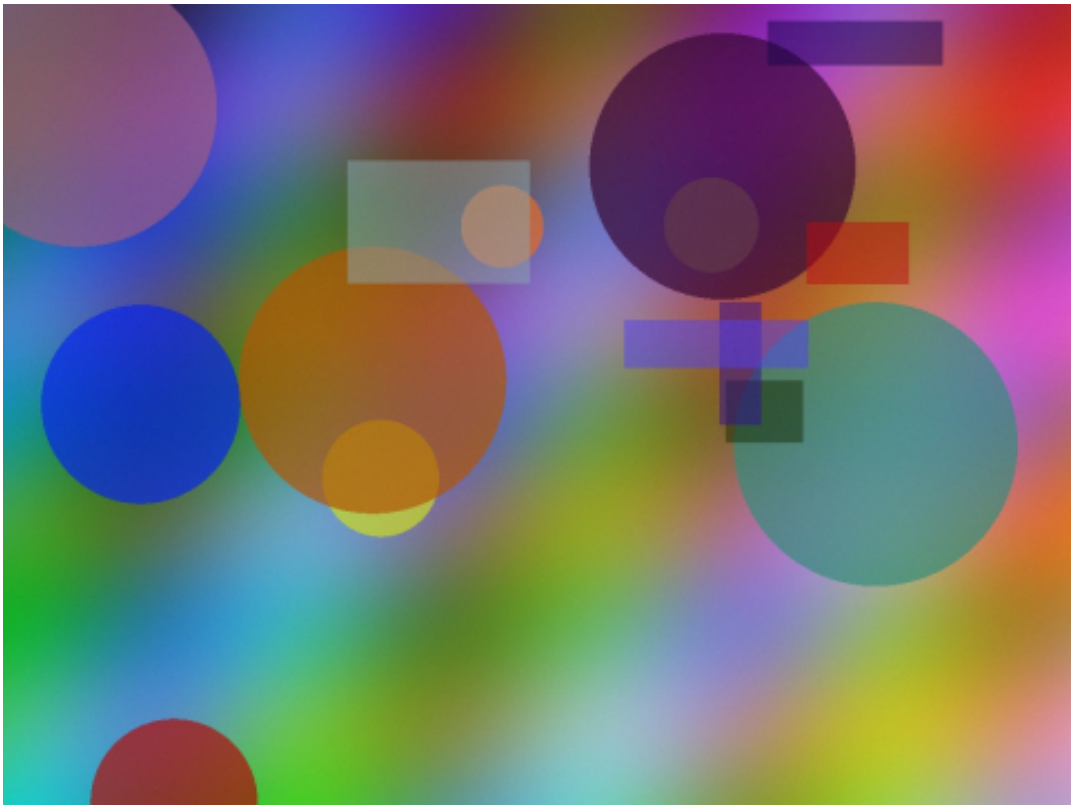
What it shows	No entry of the fingerprint databases matches these tables (SHA-256 cfe3218ae132279888e3a2b4c6949ea5cfa8df2c92a42e1dac9597d5b622b545).
What it does not prove	The database covers few devices and apps; an unknown table says nothing about authenticity. Add known originals with `fingerprints learn`.

Image analysis

Status	Completed
Format	JPEG
Dimensions (pixels)	1200 × 900

Original image (decoded preview)

Lossless PNG of the decoded pixels, exactly as the analyses saw them (stored orientation; EXIF rotation not applied). The maps below have the same size and can be laid over it.



maps/original.png (1200x900) — reduced copy for print; the full-resolution file is in the item folder

Metadata summary

Key fields only; metadata.json in the item folder holds every tag read. Metadata is written by the device or program that saved the file and can be edited or forged.

Tags read	26
Tags per group	Exif 13, IFD0 7, IFD1 6
GPS tags	absent
MakerNote	present
XMP packet	absent
Embedded thumbnail	present
ICC profile (SHA-256)	—

Group	Field	Value
IFD0	Make	Canon
IFD0	Model	Canon EOS R6
IFD0	DateTime	2031:05:14 10:25:02
Exif	DateTimeOriginal	2031:05:14 10:25:02
Exif	DateTimeDigitized	2031:05:14 10:25:02
Exif	OffsetTime	+02:00
Exif	OffsetTimeOriginal	+02:00
Exif	PixelXDimension	1200
Exif	PixelYDimension	900
IFD0	Orientation	1 (row 0 at top and column 0 at left)

Structure summary

Top-level segments in file order (JPEG markers, PNG chunks, RIFF chunks or HEIF boxes); structure.json holds the full tree with byte offsets.

Top-level segments	13
frame (SOF)	SOF0
chroma subsampling	4:2:0
components	3
restart interval	75
quantization tables	2
Huffman tables	4
bytes after EOI	0
comments (COM)	0

Marker / chunk / box sequence:

```
SOI · APP1/Exif · DQT ×2 · SOF0 · DHT ×4 · DRI · SOS · EOI
```

Quantization tables

Values in natural (row-major) order, as stored in the DQT segments. The IJG comparison names a standard libjpeg quality only on an exact match; otherwise the nearest quality and its distance (sum of absolute differences) are given, which is not a quality percentage.

IJG comparison	non-IJG; nearest q=90, distance 578
Database match	no entry in the loaded databases
Table hash (SHA-256)	`cfe3218ae132279888e3a2b4c6949ea5cfa8df2c92a42e1dac9597d5b622b545`

Table 0 · 8-bit · Used by components: 1

```
2   3   4   5   6   7   8   9
3   4   5   6   7   8  10  11
4   5   6   8   9  10  12  13
5   6   8   9  11  12  14  15
6   7   9  11  12  14  16  17
7   8  10  12  14  16  18  19
8  10  12  14  16  18  20  22
9  11  13  15  17  19  22  24
```

Table 1 · 8-bit · Used by components: 2, 3

```
3   5   7   9  11  13  15  17
5   7   9  12  14  16  19  21
7   9  12  15  17  20  23  25
9  12  15  18  21  24  27  30
11  14  17  21  24  27  31  34
13  16  20  24  27  31  35  38
15  19  23  27  31  35  39  43
17  21  25  30  34  38  43  47
```

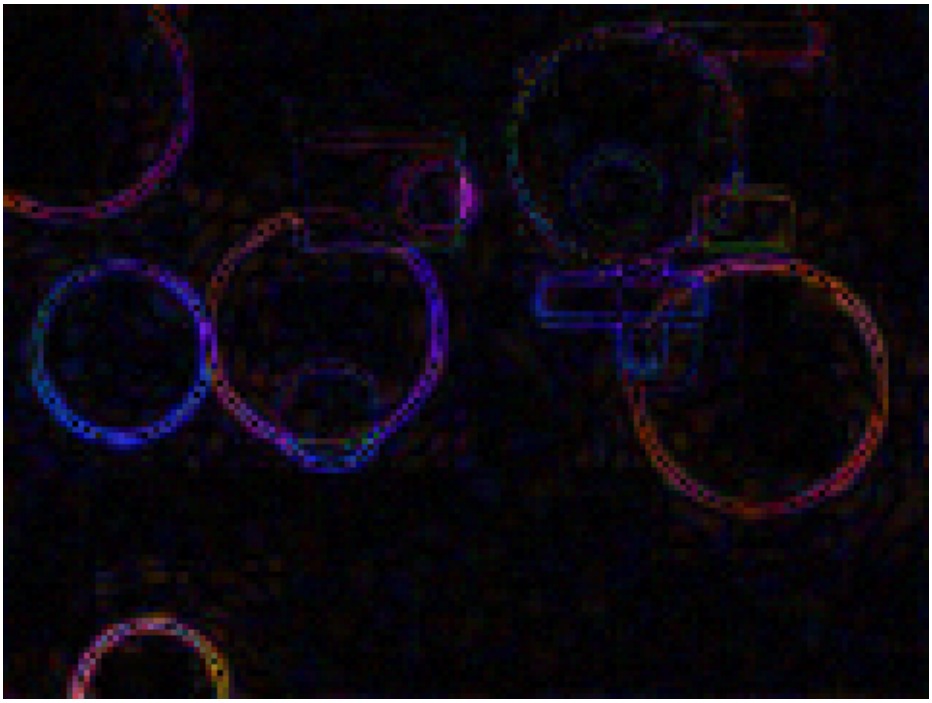
Fingerprint databases

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	0	0	loaded	`e0b0c87e4b79e5c5402e920d343436056c25bb959a5bb757a7d0519b95ab4be8`
local	—	—	—	absent	—

Analysis maps

Visual aid, requires interpretation: these maps have no score and the tool draws no conclusion from them. Bright areas are not proof of a change; authentic edges, textures and noise produce them too.

Thumbnail vs main image difference



maps/thumbnail-diff.png — reduced copy for print; the full-resolution file is in the item folder

compared_area	160x120 at 0,0 (black bars trimmed)
gain	4
largest_block_difference	9.7
mean_difference	2.6
resize	main image area-averaged to the thumbnail size
thumbnail_size	160x120
upscale	4

Copy-move shift vectors

No copy-move candidate was found (scaled or rotated copies are not detected).

Extracted files

structure.json (structure, 11002 B)	md5: f207eaa4f11b5e7af6a8cedc1fef7de2 sha256: 30b9f0583c2967a5f01942a9f8c81e1e73c0bd79557142753b7625bc522be0d7
metadata.json (metadata, 3014 B)	md5: 6bd1c3a2f050217ca9513a21d6899a41 sha256: dd02b3d88f941c9aa000c8ef0b61128db87df9de0b112822149fc53bb7f3afe2
maps/original.png (original_preview, 1181973 B)	md5: f4cf4485a057e8ff77ed992285610f45 sha256: fcc59ce43c02234e20c95faa395e8d3bca855517e91f36b694fa313226dffcc1

maps/thumbnail-diff.png (map, 69967 B)	md5: efb3932faa5a42659cb324528aab5d2b sha256: c213e7ff8db70b580bf676128a141281298b0acd7a5f8a35e668c0b9aba10596

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms, and its real type was identified from its signature. JPEG, HEIC, PNG and WebP images were then re-read in an isolated process and the bytes read were hashed again and compared with the first pass. The image analysis read the file structure and metadata, compared the JPEG quantization tables with the IJG standard and the fingerprint databases, looked for double JPEG compression, wrote the selected visual maps (error level, noise, JPEG ghosts), searched for copied-and-moved regions, compared the embedded thumbnail with the main image and, when reference images were given, compared the image with them. HEIC pixels are not decoded (structure and metadata only). Every action and every file written was recorded in a hash-chained audit log.

Audit trail

Audit entries	#23 – #31
Log folder	`R:\case\runs\rv-imgauth\20261113T232757Z-cbb6f3\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- An image without findings is not proven authentic: careful edits can leave none of the traces these checks look for, and it is impossible to prove a negative (SWGDE 18-I-001).
- Error level, noise and ghost maps are visual aids for a trained examiner; the tool gives them no score and draws no conclusion from them.
- Double JPEG compression shows the image was saved more than once; ordinary sharing and exporting do this.
- The compression-table database covers few devices and apps (built from Rakvor's own captures plus local learning); an absent or matching entry is weak evidence on its own.
- Copy-move detection compares blocks at the same scale and orientation on a downscaled image: scaled, rotated or small copies are not found.
- Error rates of these checks have not yet been measured on a documented validation corpus; the findings are signals to be weighed by the examiner.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.